ASE CHAPTER 1

T6 · Create (Diagram)

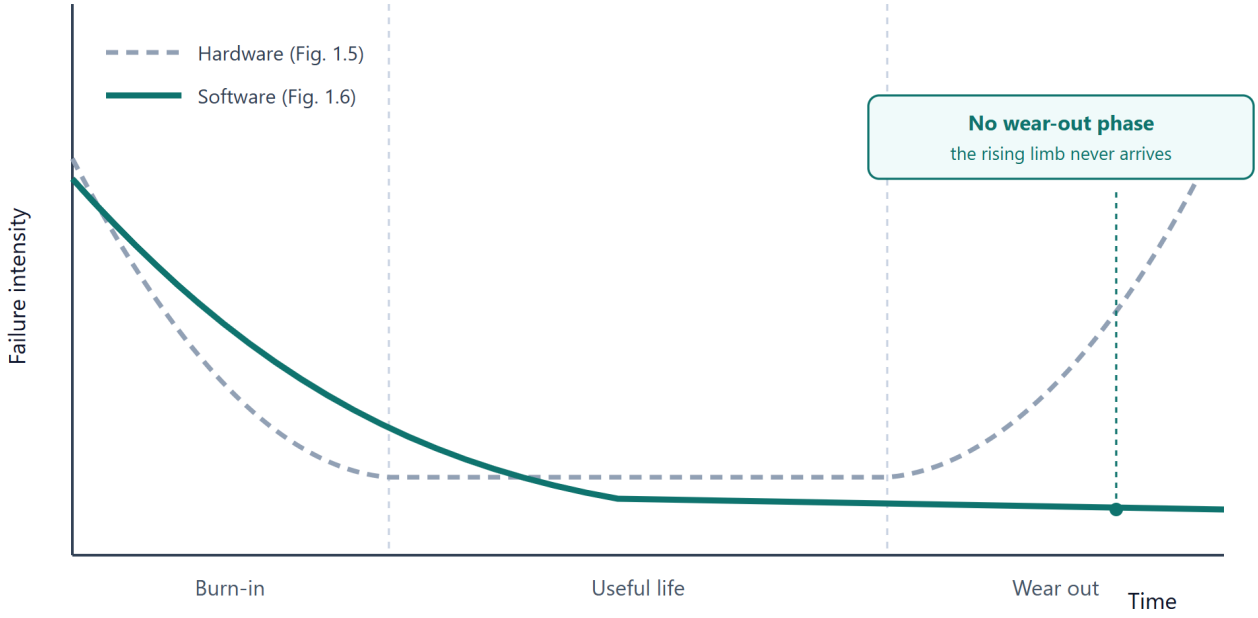
The software non-wear-out curve against the hardware bath-tub curve — answers in English, notes in Arabic

Retrieval set — T6 Create

D.1 · Draw the software failure-intensity curve against the hardware bath-tub curve, labelling the axes and the phases.

Answer

Axes: failure intensity (vertical) against time (horizontal). The hardware curve (Fig. 1.5) has three phases: high intensity during **burn-in**, a low flat **useful-life** plateau, then a **rising** limb at **wear out**. The software curve (Fig. 1.6) has **only the descending curve**: defects are found and fixed, so intensity keeps falling and the rising limb never arrives.



Failure intensity against time — hardware bath-tub (Fig. 1.5) vs software (Fig. 1.6)

Aggarwal, §1.2.4, pp.8–9; Fig. 1.5 p.8, Fig. 1.6 p.9.

D.2 · Interpret the drawing: why does the software curve never rise again?

Software does not wear out and has no wear-out phase. It **becomes obsolete when the environment it was developed for changes.**

Aggarwal, §1.2.4, pp.8–10.

Rapid self-test — recall prompts (no answers shown)

Name the three phases of the hardware curve.
What shape does the software curve take, and why?
What happens to software instead of wearing out?